

Learning Theory (Lecture 7)

COMP0078: Supervised Learning

Course Notes

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Revision notation: items marked with $\star$ are specifically listed in the revision notes file ([revision25-26-V1.pdf](#)). Non-examinable proofs (NFL proof sketch and Fundamental Theorem lower/upper bound proofs) have been removed; theorem statements are preserved.

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1 Introduction

1.1 Review and Notation

We recall the supervised learning framework introduced in the first lecture. For simplicity we consider the 2-class setting following the notation from Understanding Machine Learning [SS14].

Definition 1.1 (Probability Notation). *If $\mathcal{D}$ is a distribution over $\mathcal{Z}$ and $A \subseteq \mathcal{Z}$, then $\mathcal{D}(A)$ denotes the probability that if z is drawn from $\mathcal{D}$, then $z \in A$.*

Definition 1.2 (Expected Error^{*}). *Data is sampled IID from a distribution $\mathcal{D}$ (previously denoted P) over $\mathcal{X} \times \mathcal{Y}$ with $\mathcal{Y} = \{0, 1\}$. The **expected error** (also called **generalisation error**) of a function $h : \mathcal{X} \rightarrow \mathcal{Y}$ is*

$$L_{\mathcal{D}}(h) = \mathcal{D}(\{(x, y) : h(x) \neq y\}) = \mathbb{P}_{(x, y) \sim \mathcal{D}}[h(x) \neq y] \quad (1)$$

In our previous notation, $L_{\mathcal{D}}(h) = \mathcal{E}(h) = \int [h(x) \neq y] dP(x, y)$.

Definition 1.3 (Empirical Error^{*}). *The **empirical error** of h with respect to a dataset $S = \{(x_1, y_1), \dots, (x_m, y_m)\}$ is*

$$L_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbb{I}[h(x_i) \neq y_i] \quad (2)$$

(previously denoted $\mathcal{E}_{\text{emp}}(S, h)$).

1.2 Validation Set Bound^{*}

Theorem 1.4 (Validation Set Bound^{*}). *Select a hypothesis h . For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the random sample V of size m from $\mathcal{D}$, we have*

$$L_{\mathcal{D}}(h) \leq L_V(h) + \sqrt{\frac{\ln(1/\delta)}{2m}} \quad (3)$$

where $L_{\mathcal{D}}(h)$ and $L_V(h)$ are as in Definitions 1.2 and 1.3.

Remark 1.5 (Interpretation and Application). *The generalisation error of a function h may be bounded by its empirical error. Thus in practice we may select a predictor h based on a training set S , then we may bound its generalisation error by validating on a separate set of data V .*

We will use a concentration inequality to prove Theorem 1.4.

Lemma 1.6 (Hoeffding's Inequality). *Let $Z_1, Z_2, \dots, Z_m$ be IID Bernoulli random variables where for all i , $\mathbb{P}[Z_i = 1] = p$ and $\mathbb{P}[Z_i = 0] = 1 - p$. Let $\bar{Z} := \frac{1}{m} \sum_{i=1}^m Z_i$. Then for any $\epsilon > 0$,*

$$\mathbb{P}[\bar{Z} > p + \epsilon] \leq \exp(-2m\epsilon^2) \quad (4)$$

$$\mathbb{P}[\bar{Z} < p - \epsilon] \leq \exp(-2m\epsilon^2) \quad (5)$$

Remark 1.7 (Concentration Inequalities). *The law of large numbers states that the empirical average of an IID sequence of random variables $\frac{1}{m} \sum_{i=1}^m Z_i$ will converge to the mean $\mu = \mathbb{E}[Z_i]$. Concentration inequalities quantify how the empirical average deviates from the (true) mean when m is finite.*

Proof of Theorem 1.4. Given h , we will bound

$$L_{\mathcal{D}}(h) - L_V(h) = \mathbb{P}_{(x, y) \sim \mathcal{D}}[h(x) \neq y] - \frac{1}{m} \sum_{i=1}^m \mathbb{I}[h(x_i) \neq y_i] \quad (6)$$

1. Define $Z_i := \mathbb{1}[h(x_i) \neq y_i]$.
2. $Z_1, \dots, Z_m$ are statistically independent.
3. For all i : $\mathbb{P}[Z_i = 1] = L_{\mathcal{D}}(h) = \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y]$.
4. Applying Hoeffding's inequality (Lemma 1.6, specifically (4)) gives

$$\mathbb{P}[L_{\mathcal{D}}(h) - L_V(h) > \epsilon] \leq \exp(-2\epsilon^2 m) \quad (7)$$

5. Set $\delta := \exp(-2\epsilon^2 m)$ and solving for ϵ gives $\epsilon = \sqrt{\frac{\ln(1/\delta)}{2m}}$.
6. Note: if we use both the upper and lower bounds in Lemma 1.6 (i.e., both (4) and (5)), then alternately

$$|L_{\mathcal{D}}(h) - L_V(h)| \leq \sqrt{\frac{\ln(2/\delta)}{2m}} \quad (8)$$

□

1.3 Observations on the Validation Bound

The validation bound (Theorem 1.4) is a nice result with minimal assumptions. However, it has drawbacks:

1. The validation bound gives a way to estimate a confidence interval for our generalisation error. However, the data V used for validating cannot have been used for training, thus it is expensive in data.
2. Often, we have only few examples. Can we choose a model and assess its expected error directly (without splitting the training data)?
3. Rather than only derive confidence bounds about a predictor h , we would like to analyse predictors derived from machine learning algorithms – thus additionally deriving insights into ML algorithms.
4. Learning theory studies conditions which ensure the predictivity of a learning algorithm:
 - The expected error is close to the empirical error
 - The expected error decreases when the number of samples increases
5. In this lecture we will limit ourselves to the study of Empirical Risk Minimisation.

1.4 Statistical Learning Theory

- Basic approach of Statistical Learning Theory (SLT) is to view learning from a statistical viewpoint.
- Aim of any theory is to model real/artificial phenomena so that we can better understand/predict/exploit them.
- SLT is just one approach to understanding/predicting/exploiting learning systems. Others include Bayesian inference, inductive inference, statistical physics, and traditional statistical analysis.
- Each theory makes assumptions about the phenomenon of learning and based on these derives predictions of behaviour as well as algorithms that aim at optimising the predictions.

- Each theory has strengths and weaknesses – the better it captures the details of real world experience, the better the theory and the better the chances of it making accurate predictions and driving good algorithms.

1.5 General Statistical Considerations

- Statistical models (not including Bayesian) begin with an assumption that the data is generated by an underlying distribution $\mathcal{D}$, typically not given explicitly to the learner.
- Usually the distribution subsumes the processes of the natural/artificial world that we are studying.
- Rather than accessing the distribution directly, statistical learning typically assumes that we are given a **training sample** or **training set**

$$S = \{(x_1, y_1), \dots, (x_m, y_m)\} \quad (9)$$

generated identically and independently (i.i.d.) according to the distribution $\mathcal{D}$.

1.6 Generalisation of a Learner

Definition 1.8 (Learning Algorithm). *Assume that we have a learning algorithm $\mathcal{A}$ that chooses a hypothesis (function) $\mathcal{A}_H(S)$ from a hypothesis (function) space H in response to the training set S .*

We are interested in the study of $L_{\mathcal{D}}(\mathcal{A}_H(S))$ (Definition 1.2), i.e., the generalisation error of our algorithm $\mathcal{A}_H(S)$, which is a random variable dependent on the draw of S .

Definition 1.9 (Empirical Risk Minimization*). *In particular, we will study **empirical risk minimization***

$$ERM_H(S) := \arg \min_{h \in H} L_S(h) \quad (10)$$

where $L_S(h)$ is the empirical error (Definition 1.3). *Note: There may be many possible empirical risk minimisers; we assume $ERM_H(S)$ outputs an arbitrary one.*

1.7 Bayes Risk and Consistency

- Traditional statistics has concentrated on analysing

$$\lim_{m \rightarrow \infty} \mathbb{E}_{S_m} [L_{\mathcal{D}}(\mathcal{A}(S_m))] \quad (11)$$

where S_m denotes a training set of size m .

- Consistency of a classification algorithm means convergence to the function with the lowest possible risk, often referred to as the **Bayes risk**.

1.8 Expected versus Confident Bounds*

- For a finite sample, the generalisation $L_{\mathcal{D}}(\mathcal{A}(S_m))$ has a distribution depending on the algorithm, function class, and sample size m .
- Traditional statistics has concentrated on the mean of this distribution – but this quantity can be misleading (e.g., for low fold cross-validation).
- Statistical learning theory has preferred to analyse the **tail of the distribution**, finding a bound which holds with high probability.

- This looks like a statistical test – significant at a 1% confidence means that the chances of the conclusion not being true are less than 1% over random samples of that size.
- This is the source of the acronym **PAC: Probably Approximately Correct**. The ‘confidence’ parameter δ is the probability that we have been misled by the training set.

2 PAC Learning*

2.1 Realizability Assumption*

Definition 2.1 (Realizability Assumption*). *We assume that there exists some true function f^* so that for all $x \in \mathcal{X}$ we have $f^*(x) = y$ (i.e., there exists a classifier with zero error).*

Remark 2.2 (Implication of Realizability). *An implication of this assumption is that we can treat $\mathcal{D}$ now as only a distribution over $\mathcal{X}$ (as opposed to $\mathcal{X} \times \mathcal{Y}$).*

To indicate the dependence on f^* , we will now use the notation

$$L_{\mathcal{D},f^*}(h) = \mathbb{P}_{x \sim \mathcal{D}}[h(x) \neq f^*(x)] \quad (12)$$

Question: Can we find an algorithm $\mathcal{A}$ so that $h = \mathcal{A}(S)$ such that $L_{\mathcal{D},f^*}(h)$ is small?

2.2 Can Only Be Approximately Correct

Lemma 2.3 (Impossibility of Zero Error). *We cannot hope to find h such that $L_{\mathcal{D},f^*}(h) = 0$.*

Proof. For every $\epsilon \in (0, 1)$, take $\mathcal{X} = \{x_1, x_2\}$ and $\mathcal{D}(\{x_1\}) = 1 - \epsilon$, $\mathcal{D}(\{x_2\}) = \epsilon$.

The probability of not seeing x_2 at all among m i.i.d. examples is

$$(1 - \epsilon)^m \approx e^{-\epsilon m} \quad (13)$$

So if $\epsilon \ll 1/m$, we’re likely not to see x_2 at all, but then we cannot know its label.

Relaxation: We’d be happy with $L_{\mathcal{D},f^*}(h) \leq \epsilon$, where ϵ is user-specified. $\square$

2.3 Can Only Be Probably Correct

- Recall that the input to the learner is randomly generated.
- There’s always a (very small) chance to see the same example again and again.

Lemma 2.4 (Impossibility of Deterministic Guarantee). *No algorithm can guarantee $L_{\mathcal{D},f^*}(h) \leq \epsilon$ for sure.*

Relaxation: We’d allow the algorithm to fail with probability δ , where $\delta \in (0, 1)$ is user-specified. Here, the probability is over the random choice of examples.

2.4 Probably Approximately Correct (PAC) Learning*

- The learner doesn’t know $\mathcal{D}$ and f^* .
- The learner receives accuracy parameter ϵ and confidence parameter δ .
- The learner can ask for training data, S , containing $m(\epsilon, \delta)$ examples (that is, the number of examples can depend on the value of ϵ and δ , but it cannot depend on $\mathcal{D}$ or f^*).
- Learner should output a hypothesis h such that with probability of at least $1 - \delta$ it holds that $L_{\mathcal{D},f^*}(h) \leq \epsilon$.
- That is, the learner should be **Probably** (with probability at least $1 - \delta$) **Approximately** (up to accuracy ϵ) **Correct** (by Lemma 2.3 and Lemma 2.4, these relaxations are necessary).

2.5 No Free Lunch*

Theorem 2.5 (No Free Lunch*). Suppose that $|\mathcal{X}| = \infty$. Fix $\delta \in (0, 1)$, $\epsilon < 1/2$. For every learner $\mathcal{A}$ and training set size m , there exists $\mathcal{D}$, f^* such that with probability of at least δ over the generation of a training data S of m examples, it holds that

$$L_{\mathcal{D}, f^*}(\mathcal{A}(S)) \geq \epsilon \quad (14)$$

Remark 2.6 (Interpretation of No Free Lunch). Note that $L_{\mathcal{D}, f^*}(\text{random guess}) = 1/2$, so the theorem states that you cannot be better than a random guess. This is a stronger result in expectation than given in Lecture 1 (see Problem 5.3 in [SS14]).

2.6 Prior Knowledge

- Give more knowledge to the learner: the target f^* comes from some hypothesis class $H \subseteq \mathcal{Y}^{\mathcal{X}}$.
- The learner knows H .
- Is it possible to PAC learn now?
- Of course, the answer depends on H , since the No Free Lunch theorem (Theorem 2.5) tells us that for $H = \mathcal{Y}^{\mathcal{X}}$ the problem is not solvable.

3 PAC Learning of Finite Hypothesis Classes*

3.1 Learning Finite Classes

Assume that H is a finite hypothesis class. For example, H could be all the functions from $\mathcal{X}$ to $\mathcal{Y}$ that can be implemented using a Python program of length at most b .

Definition 3.1 (Consistent Learning Rule / ERM). Given H and $S = \{(x_1, y_1), \dots, (x_m, y_m)\}$, output any $h \in H$ such that $\forall i, y_i = h(x_i)$.

This is also called *Empirical Risk Minimization* (Definition 1.9):

$$ERM_H(S) = \arg \min_{h \in H} L_S(h) \quad (15)$$

where the empirical risk is $L_S(h) = \frac{1}{m} |\{i : h(x_i) \neq y_i\}|$.

3.2 Main Theorem for Finite Classes

Theorem 3.2 (PAC Bound for Finite Classes*). Fix $\epsilon, \delta \in (0, 1)$. If

$$m \geq \frac{\log(|H|/\delta)}{\epsilon} \quad (16)$$

then for every $\mathcal{D}$, f^* , with probability of at least $1 - \delta$ (with respect to the randomly sampled training set S of size m),

$$L_{\mathcal{D}, f^*}(ERM_H(S)) \leq \epsilon \quad (17)$$

Corollary 3.3 (Interpretation of Finite PAC Bound*). By rearranging Theorem 3.2, we have

$$L_{\mathcal{D}, f^*}(ERM_H(S)) \leq \frac{\log |H| + \log(1/\delta)}{m} \quad (18)$$

Thus ERM in the realizable case: generalisation error decreases linearly in the number of samples m and increases logarithmically in the size of the hypothesis class.

3.3 Proof of Theorem 3.2

Let $S|_x = (x_1, \dots, x_m)$ be the instances of the training set. We would like to prove:

$$\mathcal{D}^m(\{S|_x : L_{\mathcal{D}, f^*}(\text{ERM}_H(S)) > \epsilon\}) \leq \delta \quad (19)$$

Definition 3.4 (Bad Hypotheses). *Let H_B be the set of “bad” hypotheses:*

$$H_B = \{h \in H : L_{\mathcal{D}, f^*}(h) > \epsilon\} \quad (20)$$

Definition 3.5 (Misleading Samples). *Let M be the set of “misleading” samples:*

$$M = \{S|_x : \exists h \in H_B, L_S(h) = 0\} \quad (21)$$

Key Observation:

$$\{S|_x : L_{\mathcal{D}, f^*}(\text{ERM}_H(S)) > \epsilon\} \subseteq M = \bigcup_{h \in H_B} \{S|_x : L_S(h) = 0\} \quad (22)$$

Lemma 3.6 (Union Bound). *For any two sets A, B and a distribution $\mathcal{D}$, we have*

$$\mathcal{D}(A \cup B) \leq \mathcal{D}(A) + \mathcal{D}(B) \quad (23)$$

Proof of Theorem 3.2. Using the inclusion (22) and the union bound (Lemma 3.6):

$$\begin{aligned} \mathcal{D}^m(\{S|_x : L_{\mathcal{D}, f^*}(\text{ERM}_H(S)) > \epsilon\}) &\leq \sum_{h \in H_B} \mathcal{D}^m(\{S|_x : L_S(h) = 0\}) \\ &\leq |H_B| \max_{h \in H_B} \mathcal{D}^m(\{S|_x : L_S(h) = 0\}) \end{aligned} \quad (24)$$

Computing the probability:

Under the realizability assumption (Definition 2.1), observe:

$$\mathcal{D}^m(\{S|_x : L_S(h) = 0\}) = (1 - L_{\mathcal{D}, f^*}(h))^m \quad (25)$$

If $h \in H_B$ (Definition 3.4), then $L_{\mathcal{D}, f^*}(h) > \epsilon$, and therefore

$$\mathcal{D}^m(\{S|_x : L_S(h) = 0\}) < (1 - \epsilon)^m \quad (26)$$

We have shown:

$$\mathcal{D}^m(\{S|_x : L_{\mathcal{D}, f^*}(\text{ERM}_H(S)) > \epsilon\}) < |H_B|(1 - \epsilon)^m \quad (27)$$

Finally, using Lemma 3.7 with $1 - \epsilon \leq e^{-\epsilon}$ and $|H_B| \leq |H|$, we conclude:

$$\mathcal{D}^m(\{S|_x : L_{\mathcal{D}, f^*}(\text{ERM}_H(S)) > \epsilon\}) < |H|e^{-\epsilon m} \quad (28)$$

The right-hand side would be $\leq \delta$ if $m \geq \frac{\log(|H|/\delta)}{\epsilon}$. $\square$

Lemma 3.7 (Exponential Inequality). *For all $\epsilon \in \mathbb{R}$, $1 - \epsilon \leq e^{-\epsilon}$.*

3.4 Illustrating the Use of the Union Bound

Each point represents a possible sample $S|_x$. Each colored oval represents misleading samples (Definition 3.5) for some $h \in H_B$ (Definition 3.4). The probability mass of each such oval is at most $(1 - \epsilon)^m$. But the algorithm might err if it samples $S|_x$ from any of these ovals. The union bound (Lemma 3.6) allows us to sum over all such ovals.

4 The Fundamental Theorem of Learning Theory

4.1 PAC Learnability

Definition 4.1 (PAC Learnability). A hypothesis class H is **PAC learnable** if there exists a function $m_H : (0, 1)^2 \rightarrow \mathbb{N}$ and a learning algorithm with the following property:

For every $\epsilon, \delta \in (0, 1)$, for every distribution $\mathcal{D}$ over $\mathcal{X}$, and for every labeling function $f^* : \mathcal{X} \rightarrow \{0, 1\}$, when running the learning algorithm on $m \geq m_H(\epsilon, \delta)$ i.i.d. examples generated by $\mathcal{D}$ and labeled by f^* , the algorithm returns a hypothesis h such that, with probability of at least $1 - \delta$ (over the choice of the examples),

$$L_{\mathcal{D}, f^*}(h) \leq \epsilon \quad (29)$$

The function m_H is called the **sample complexity** of learning H .

Corollary 4.2 (Finite Classes are PAC Learnable*). Let H be a finite hypothesis class. Then:

- H is PAC learnable (Definition 4.1) with sample complexity $m_H(\epsilon, \delta) \leq \frac{\log(|H|/\delta)}{\epsilon}$
- This sample complexity is obtained by using the ERM_H learning rule (Definition 1.9)

Proof. Follows directly from Theorem 3.2. □

4.2 Questions About Learnability

- What about infinite hypothesis classes?
- What is the sample complexity of a given class?
- Is there a generic learning algorithm that achieves the optimal sample complexity?

Answer: The fundamental theorem of statistical learning (Theorem 4.15) states:

- The sample complexity is characterized by the **VC dimension** (Definition 4.6)
- The ERM learning rule is a generic (near) optimal learner

4.3 The VC Dimension – Motivation*

- Suppose we got a training set $S = \{(x_1, y_1), \dots, (x_m, y_m)\}$.
- We try to explain the labels using a hypothesis from H .
- Then, oops, the labels we received are incorrect and we get the same instances with different labels: $S' = \{(x_1, y'_1), \dots, (x_m, y'_m)\}$.
- We again try to explain the labels using a hypothesis from H .
- If this works for us, no matter what the labels are, then something is fishy...
- Formally, if H allows all functions over some set C of size m , then based on the No Free Lunch theorem (Theorem 2.5), we cannot learn from, say, $m/2$ examples.

4.4 The VC Dimension – Formal Definition*

Definition 4.3 (Restriction of H to C). Let $C = \{x_1, \dots, x_{|C|}\} \subset \mathcal{X}$. Let H_C be the restriction of H to C , namely,

$$H_C = \{h_C : h \in H\} \quad (30)$$

where $h_C : C \rightarrow \{-1, 1\}$ is such that $h_C(x_i) = h(x_i)$ for every $x_i \in C$.

Remark 4.4 (Sign Vector Representation). We can represent each h_C as the vector $(h(x_1), \dots, h(x_{|C|})) \in \{-1, 1\}^{|C|}$. Therefore: $|H_C| \leq 2^{|C|}$.

Definition 4.5 (Shattering*). We say that H **shatters** C if $|H_C| = 2^{|C|}$ (where H_C is as in Definition 4.3).

Definition 4.6 (VC Dimension*). The **VC dimension** of H is

$$VCdim(H) = \sup\{|C| : H \text{ shatters } C\} \quad (31)$$

That is, the VC dimension is the maximal size of a set C such that H gives no prior knowledge with respect to C (in the sense of Definition 4.5).

4.5 VC Dimension Illustrated

We may visualize a hypothesis space H as a sign matrix (potentially infinite), with rows representing hypotheses and columns representing instances.

The VC dimension $VCdim(H)$ is determined by the maximal $2^{VCdim(H)} \times VCdim(H)$ submatrix in which each row is distinct, i.e., all $2^{VCdim(H)}$ possible sign patterns of length $VCdim(H)$ are present.

4.6 Computing VC Dimension*

To show that $VCdim(H) = d$ (Definition 4.6), we need to show that:

1. There exists a set C of size d which is shattered by H (Definition 4.5).
2. Every set C of size $d + 1$ is not shattered by H .

4.7 Examples of VC Dimension*

Example 4.7 (Threshold Functions). Let $\mathcal{X} = \mathbb{R}$ and $H = \{x \mapsto \text{sign}(x - \theta) : \theta \in \mathbb{R}\}$ where

$$\text{sign}(z) := \begin{cases} 1 & z \geq 0 \\ -1 & z < 0 \end{cases} \quad (32)$$

Claim: $VCdim(H) = 1$.

Proof:

- Show that $\{0\}$ is shattered (Definition 4.5): For any labeling of the single point 0, we can choose θ appropriately ($\theta < 0$ for label +1, $\theta > 0$ for label -1).
- Show that any two points cannot be shattered: Take two points $a < b$. We cannot achieve the labeling $(+1, -1)$ on (a, b) because any threshold function is monotone.

Example 4.8 (Intervals). Let $\mathcal{X} = \mathbb{R}$ and $H = \{h_{a,b} : a < b \in \mathbb{R}\}$, where

$$h_{a,b}(x) = \begin{cases} 1 & x \in [a, b] \\ -1 & \text{otherwise} \end{cases} \quad (33)$$

Claim: $VCdim(H) = 2$.

Proof:

- Show that $\{0, 1\}$ is shattered: All four labelings can be achieved by choosing appropriate intervals.
- Show that any three points cannot be shattered: For three points $a < b < c$, we cannot achieve the labeling $(+1, -1, +1)$ because an interval is a contiguous region.

Example 4.9 (Axis-Aligned Rectangles). Let $\mathcal{X} = \mathbb{R}^2$ and $H = \{h_{(a_1, a_2, b_1, b_2)} : a_1 < a_2 \text{ and } b_1 < b_2\}$, where

$$h_{(a_1, a_2, b_1, b_2)}(x_1, x_2) = \begin{cases} 1 & x_1 \in [a_1, a_2] \text{ and } x_2 \in [b_1, b_2] \\ -1 & \text{otherwise} \end{cases} \quad (34)$$

Claim: $\text{VCdim}(H) = 4$.

Proof: Four points at the corners of a square can be shattered (Definition 4.5), but five points cannot (consider the point in the middle of four corner points).

Example 4.10 (Finite Classes). For any finite hypothesis class H :

$$\text{VCdim}(H) \leq \log_2(|H|) \quad (35)$$

There can be an arbitrary gap between $\text{VCdim}(H)$ and $\log_2(|H|)$.

Example 4.11 (Halfspaces). Let $\mathcal{X} = \mathbb{R}^n$ and $H = \{x \mapsto \text{sign}(\langle w, x \rangle) : w \in \mathbb{R}^n\}$.

Claim: $\text{VCdim}(H) = n$.

Proof:

- Show that $\{\eta_1, \dots, \eta_n\}$ (standard basis vectors) is shattered (Definition 4.5).
- Show that any $n + 1$ points cannot be shattered (follows from linear algebra: any $n + 1$ points in $\mathbb{R}^n$ are linearly dependent).

Example 4.12 (Large Margin Halfspaces*). Define the inner product space of bounded square summable sequences

$$\ell_2 := \left\{ x \in \mathbb{R}^\infty : \sum_{i=1}^{\infty} x_i^2 < \infty \right\} \quad (36)$$

with inner product $\langle x, x' \rangle = \sum_{i=1}^{\infty} x_i x'_i$.

Given $\mathcal{X} \subset \ell_2$ and $\Lambda \in (0, \infty)$, define

$$H_{\mathcal{X}, \Lambda} = \{x \mapsto \text{sign}(\langle w, x \rangle) : x \in \mathcal{X}, w \in \ell_2, \|w\| \leq \Lambda, |\langle w, x \rangle| \geq 1\} \quad (37)$$

Observe that $1/\|w\|$ is the margin.

Theorem 4.13 (VC Dimension of Large Margin Halfspaces*). For $H_{\mathcal{X}, \Lambda}$ as in (37),

$$\text{VCdim}(H_{\mathcal{X}, \Lambda}) \leq \Lambda^2 \max_{x \in \mathcal{X}} \|x\|^2 \quad (38)$$

Remark 4.14 (Tightness of Large Margin Bound). Furthermore, for every $\Lambda \in (0, \infty)$, there exists an $\mathcal{X}$ such that

$$\text{VCdim}(H_{\mathcal{X}, \Lambda}) = \lfloor \Lambda^2 \max_{x \in \mathcal{X}} \|x\|^2 \rfloor \quad (39)$$

4.8 The Fundamental Theorem*

Theorem 4.15 (The Fundamental Theorem of Statistical Learning*). Let H be a hypothesis class of binary classifiers. Then there are absolute constants C_1, C_2 such that the sample complexity of PAC learning H (Definition 4.1) is

$$C_1 \frac{\text{VCdim}(H) + \log(1/\delta)}{\epsilon} \leq m_H(\epsilon, \delta) \leq C_2 \frac{\text{VCdim}(H) \log(1/\epsilon) + \log(1/\delta)}{\epsilon} \quad (40)$$

where $\text{VCdim}(H)$ is as in Definition 4.6. Furthermore, this sample complexity is achieved by the ERM learning rule (Definition 1.9).

4.9 Growth Function

Definition 4.16 (Growth Function). *The **growth function** $\Pi_H(m)$ is the maximum cardinality of the set of functions in H when restricted to m points (“columns”, cf. Definition 4.3):*

$$\Pi_H(m) := \max_{C \subset \mathcal{X}} \{|H_C| : |C| = m\} \quad (41)$$

Thus $\text{VCdim}(H) = \max\{m : \Pi_H(m) = 2^m\}$ (cf. Definition 4.6).

Lemma 4.17 (Sauer–Shelah–Perles–Vapnik–Chervonenkis). *Let H be a hypothesis class with $d = \text{VCdim}(H) < \infty$ (Definition 4.6). Then the growth function (Definition 4.16) may be bounded as*

$$\Pi_H(m) = \begin{cases} 2^m & m \leq d \\ \leq \sum_{i=0}^d \binom{m}{i} \leq \left(\frac{em}{d}\right)^d & m > d \end{cases} \quad (42)$$

Remark 4.18 (Interpretation of Sauer’s Lemma). *The VC dimension is the point where the growth function rapidly decreases. That is, where we now have polynomial growth in the number of distinct hypotheses as opposed to exponential. It is precisely this slowed growth that allows the bound in Theorem 4.15 to be proven.*

5 Agnostic PAC Learning*

5.1 Relaxing the Realizability Assumption

So far we assumed that labels are generated by some $f^* \in H$ (the realizability assumption from Definition 2.1). This assumption may be too strong.

We relax the realizability assumption by replacing the “target labeling function” with a more flexible notion: a data-labels generating distribution.

We now return to the assumption that $\mathcal{D}$ is a distribution over $\mathcal{X} \times \mathcal{Y}$.

We now use $L_{\mathcal{D}}(h)$ (as in Definition 1.2, as opposed to $L_{\mathcal{D},f^*}(h)$ in (12)):

$$L_{\mathcal{D}}(h) := \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y] = \mathcal{D}(\{(x,y) : h(x) \neq y\}) \quad (43)$$

We redefine the “approximately correct” notion to:

$$L_{\mathcal{D}}(\mathcal{A}(S)) \leq \min_{h \in H} L_{\mathcal{D}}(h) + \epsilon \quad (44)$$

Remark 5.1 (Extension to Infinite Classes). *We will prove the result for finite H . However, the proof for infinite hypothesis classes with finite $\text{VCdim}(H)$ (Definition 4.6) carries over in the same way in the agnostic setting.*

5.2 The General PAC Learning Problem

We wish to **Probably Approximately Solve**:

$$\min_{h \in H} L_{\mathcal{D}}(h) \quad \text{where} \quad L_{\mathcal{D}}(h) = \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y] \quad (45)$$

- Learner knows H
- Learner receives accuracy parameter ϵ and confidence parameter δ
- Learner can decide on training set size m based on ϵ, δ
- Learner doesn’t know $\mathcal{D}$ but can sample $S \sim \mathcal{D}^m$
- Using S , the learner outputs some hypothesis $\mathcal{A}(S)$
- We want that with probability of at least $1 - \delta$ over the choice of S , the following holds:

$$L_{\mathcal{D}}(\mathcal{A}(S)) \leq \min_{h \in H} L_{\mathcal{D}}(h) + \epsilon \quad (46)$$

5.3 Formal Definition

Definition 5.2 (Agnostic PAC Learnability*). A hypothesis class H is **agnostic PAC learnable** if there exists a function $m_H : (0, 1)^2 \rightarrow \mathbb{N}$ and a learning algorithm $\mathcal{A}$ with the following property:

For every $\epsilon, \delta \in (0, 1)$, $m \geq m_H(\epsilon, \delta)$, and distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$,

$$\mathcal{D}^m \left(\left\{ S \in (\mathcal{X} \times \mathcal{Y})^m : L_{\mathcal{D}}(\mathcal{A}(S)) \leq \min_{h \in H} L_{\mathcal{D}}(h) + \epsilon \right\} \right) \geq 1 - \delta \quad (47)$$

6 Learning via Uniform Convergence

6.1 Representative Sample

Definition 6.1 (ϵ -Representative Sample). A training set S is called **ϵ -representative** if

$$\forall h \in H, \quad |L_S(h) - L_{\mathcal{D}}(h)| \leq \epsilon \quad (48)$$

Lemma 6.2 (ERM on Representative Samples). Assume that a training set S is $\frac{\epsilon}{2}$ -representative (Definition 6.1). Then any output of $\text{ERM}_H(S)$ (Definition 1.9), namely any $h_S \in \arg \min_{h \in H} L_S(h)$, satisfies

$$L_{\mathcal{D}}(h_S) \leq \min_{h \in H} L_{\mathcal{D}}(h) + \epsilon \quad (49)$$

Proof. For every $h \in H$,

$$L_{\mathcal{D}}(h_S) \leq L_S(h_S) + \frac{\epsilon}{2} \quad (50)$$

$$\leq L_S(h) + \frac{\epsilon}{2} \quad (\text{since } h_S \text{ minimizes } L_S) \quad (51)$$

$$\leq L_{\mathcal{D}}(h) + \frac{\epsilon}{2} + \frac{\epsilon}{2} \quad (52)$$

$$= L_{\mathcal{D}}(h) + \epsilon$$

where (50) and (52) use the $\frac{\epsilon}{2}$ -representativeness of S (Definition 6.1), and (51) uses the definition of ERM (Definition 1.9). Taking the minimum over $h \in H$ completes the proof. $\square$

6.2 Uniform Convergence

Definition 6.3 (Uniform Convergence). H has the **uniform convergence property** if there exists a function $m_H^{UC} : (0, 1)^2 \rightarrow \mathbb{N}$ such that for every $\epsilon, \delta \in (0, 1)$ and every distribution $\mathcal{D}$,

$$\mathcal{D}^m(\{S \in \mathcal{Z}^m : S \text{ is } \epsilon\text{-representative}\}) \geq 1 - \delta \quad (53)$$

with $m \geq m_H^{UC}(\epsilon, \delta)$, where ϵ -representative is as in Definition 6.1.

Corollary 6.4 (Uniform Convergence Implies Learnability). • If H has the uniform convergence property (Definition 6.3) with a function m_H^{UC} , then H is agnostically PAC learnable (Definition 5.2) with sample complexity

$$m_H(\epsilon, \delta) \leq m_H^{UC}(\epsilon/2, \delta) \quad (54)$$

- Furthermore, in that case, the ERM_H paradigm (Definition 1.9) is a successful agnostic PAC learner for H .

Proof. Follows immediately from Lemma 6.2: if S is $\frac{\epsilon}{2}$ -representative then $\text{ERM}_H(S)$ achieves the agnostic PAC guarantee (44). $\square$

6.3 Finite Classes are Agnostic PAC Learnable

Theorem 6.5 (Agnostic PAC Bound for Finite Classes*). *Assume H is finite. Then H is agnostically PAC learnable (Definition 5.2) using the ERM_H algorithm (Definition 1.9) with sample complexity*

$$m_H(\epsilon, \delta) \leq \left\lceil \frac{2 \log(2|H|/\delta)}{\epsilon^2} \right\rceil \quad (55)$$

Corollary 6.6 (Interpretation of Agnostic Bound). *By rearranging Theorem 6.5, we have*

$$L_{\mathcal{D}}(ERM_H(S)) \leq \min_{h \in H} L_{\mathcal{D}}(h) + 2\sqrt{\frac{\log |H| + \log(2/\delta)}{2m}} \quad (56)$$

Comparing to the realizable case (Theorem 3.2, Corollary 3.3), the generalisation error now decreases at rate $O(1/\sqrt{m})$ as opposed to $O(1/m)$ linearly.

6.4 Proof of Theorem 6.5

Proof of Theorem 6.5. By Corollary 6.4, it suffices to show that H has the uniform convergence property (Definition 6.3) with

$$m_H^{UC}(\epsilon, \delta) \leq \left\lceil \frac{\log(2|H|/\delta)}{2\epsilon^2} \right\rceil \quad (57)$$

To show uniform convergence, we need:

$$\mathcal{D}^m(\{S : \exists h \in H, |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\}) < \delta \quad (58)$$

Using the union bound (Lemma 3.6), compare to the proof of Theorem 3.2 (specifically (24)):

$$\begin{aligned} & \mathcal{D}^m(\{S : \exists h \in H, |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\}) \\ &= \mathcal{D}^m\left(\bigcup_{h \in H} \{S : |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\}\right) \\ &\leq \sum_{h \in H} \mathcal{D}^m(\{S : |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\}) \end{aligned} \quad (59)$$

This uses the same argument as in the validation set bound proof (Theorem 1.4):

Recall: $L_{\mathcal{D}}(h) = \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y]$ and $L_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}[h(x_i) \neq y_i]$.

Define $Z_i := \mathbb{1}[h(x_i) \neq y_i]$. Then $Z_1, \dots, Z_m$ are statistically independent.

By Hoeffding's Inequality (Lemma 1.6), using both the upper tail (4) and the lower tail (5), for any $\epsilon > 0$:

$$\mathbb{P}[|L_S(h) - L_{\mathcal{D}}(h)| > \epsilon] \leq 2 \exp(-2m\epsilon^2) \quad (60)$$

Substituting (60) into (59):

$$\mathcal{D}^m(\{S : \exists h \in H, |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\}) \leq 2|H| \exp(-2m\epsilon^2) \quad (61)$$

So if $m \geq \frac{\log(2|H|/\delta)}{2\epsilon^2}$, then the right-hand side is at most δ as required. $\square$

6.5 Error Decomposition*

Let $h_S = \text{ERM}_H(S)$ (Definition 1.9). We can decompose the risk of h_S as:

$$L_{\mathcal{D}}(h_S) = \epsilon_{\text{app}} + \epsilon_{\text{est}} \quad (62)$$

Definition 6.7 (Approximation Error*). *The **approximation error** is*

$$\epsilon_{\text{app}} = \min_{h \in H} L_{\mathcal{D}}(h) \quad (63)$$

- How much risk do we have due to restricting to H
- Doesn't depend on S
- Decreases with the complexity (size or VC dimension, Definition 4.6) of H

Definition 6.8 (Estimation Error*). *The **estimation error** is*

$$\epsilon_{\text{est}} = L_{\mathcal{D}}(h_S) - \min_{h \in H} L_{\mathcal{D}}(h) \quad (64)$$

- Result of L_S being only an estimate of $L_{\mathcal{D}}$
- Decreases with the size of S
- Increases with the complexity of H

Remark 6.9 (Bias/Complexity Tradeoff*). *Choosing $H' \supset H$ leads to decreased ϵ_{app} (Definition 6.7) while ϵ_{est} (Definition 6.8) is increased.*

7 Summary

- PAC learning (Definition 4.1) provides guarantees on the generalisation error under worst-case distributional assumptions.
- ERM (Definition 1.9) is near optimal in this setting (Theorem 4.15).
- For many interesting problems, there are no efficient ERM algorithms. For example, linear classification under 0-1 loss, or k -literal disjunction learning.
- The fundamental theorem of statistical learning (Theorem 4.15) characterizes learnability through the VC dimension (Definition 4.6).
- The ERM learning rule is a generic (near) optimal learner.

8 Problems

1. Problem 3.3 in [SS14]
2. Prove $(1 - x) \leq e^{-x}$ for $x \in \mathbb{R}$ (cf. Lemma 3.7).
3. Problem 5.3 in [SS14] (No Free Lunch, cf. Theorem 2.5)
4. Problem 6.2 in [SS14]
5. Problem 6.7 in [SS14]
6. On page 49 of the slides, why did we formulate the VC dimension (Definition 4.6) of large margin classifiers (Theorem 4.13) with respect to a fixed input space $\mathcal{X}$ rather than, for example, $\{x \in \ell_2 : \|x\| \leq R\}$?
7. Prove the VC dimension results in Theorem 4.13 and Remark 4.14.

9 Suggested Readings

This lecture closely follows Chapters 3, 4, and 6 from Understanding Machine Learning: From Theory to Algorithms by S. Shalev-Shwartz and S. Ben-David, Cambridge University Press (2014) (Chapter 5 for No Free Lunch).

We omitted proofs for Sauer's Lemma (Lemma 4.17) and the generalisation of our results from finite hypothesis class to classes with bounded VC dimension (Definition 4.6). Sauer's Lemma is proved relatively directly in Section 6.5.1.

Extending from finite hypothesis classes to bounded VC dimension introduces the powerful technique of the double sample trick (cf. (??)). Proofs via Rademacher Complexities are in Sections 28.1 and 28.3 in [SS14].

For more direct insight showing the parallel to the finite hypothesis class case, see Maria-Florina Balcan's notes Lectures 5, 6, and 7: 8803 Machine Learning Theory 2010.